

Linear Initial-Value Problem for the Finite-Depth Rayleigh–Taylor Instability

Abstract

This note summarizes the linear initial-value formulation implemented in the latest Mathematica notebook. Two inviscid, incompressible, irrotational fluids are separated by a nominally flat interface at $z = 0$ and are bounded by rigid horizontal walls at $z = H_1$ and $z = -H_2$. The initial interface is flat, but an interfacial vertical perturbation velocity $v_0(x)$ is imposed at $t = 0$. The resulting dynamics are solved mode by mode using the finite-depth Rayleigh–Taylor dispersion relation with surface tension.

1 Geometry and assumptions

The upper fluid occupies

$$0 < z < H_1, \tag{1}$$

while the lower fluid occupies

$$-H_2 < z < 0. \tag{2}$$

The unperturbed interface is located at $z = 0$. The densities are denoted by ρ_1 for the upper fluid and ρ_2 for the lower fluid. The gravitational acceleration is g , and the surface tension coefficient is σ .

The interface displacement is denoted by

$$z = \eta(x, t), \tag{3}$$

where η is assumed sufficiently small for linearization. The rigid-wall impermeability conditions are

$$w_1(x, H_1, t) = 0, \quad w_2(x, -H_2, t) = 0, \tag{4}$$

where w_1 and w_2 are the vertical velocities in the upper and lower fluids, respectively.

The initial-value problem considered in the notebook is

$$\eta(x, 0) = 0, \quad \partial_t \eta(x, 0) = v_0(x). \tag{5}$$

Thus, the interface is initially flat, and the perturbation is introduced through the initial vertical velocity at the interface.

2 Mathematical background

This section records the mathematical ingredients used later in the note. The goal is to make clear where the modal equation, the finite-depth factors, and the verification tests come from.

2.1 Linearized potential-flow model

Each fluid is assumed inviscid, incompressible, and irrotational. Therefore the perturbation velocity in fluid j can be represented by a velocity potential

$$\mathbf{u}_j = (u_j, w_j) = \nabla \phi_j, \quad j = 1, 2. \quad (6)$$

Incompressibility gives

$$\nabla \cdot \mathbf{u}_j = 0, \quad (7)$$

and hence the potentials satisfy Laplace's equation in their respective fluid domains,

$$\partial_{xx}\phi_j + \partial_{zz}\phi_j = 0. \quad (8)$$

The rigid walls are impermeable, so the normal velocity vanishes there:

$$\partial_z \phi_1(x, H_1, t) = 0, \quad \partial_z \phi_2(x, -H_2, t) = 0. \quad (9)$$

The interface is the material surface $F(x, z, t) = z - \eta(x, t) = 0$. The kinematic condition $D_t F = 0$ gives

$$\partial_t \eta + u_j \partial_x \eta = w_j \quad \text{on } z = \eta(x, t). \quad (10)$$

After linearizing about the flat state $\eta = 0$ and neglecting products of perturbation quantities, this becomes

$$\partial_t \eta(x, t) = \partial_z \phi_1(x, 0, t) = \partial_z \phi_2(x, 0, t). \quad (11)$$

This condition is why the interface velocity used in the notebook is simply $\partial_t \eta$.

2.2 Fourier modes and finite-depth eigenfunctions

Because the governing equations, material parameters, and wall locations are independent of x , the linear problem is diagonal in Fourier space. In a periodic domain one may write

$$\eta(x, t) = \sum_{\kappa \neq 0} \hat{\eta}_\kappa(t) e^{i\kappa x}, \quad \phi_j(x, z, t) = \sum_{\kappa \neq 0} \hat{\phi}_{j,\kappa}(z, t) e^{i\kappa x}. \quad (12)$$

Here κ is a signed horizontal wavenumber. The vertical structure depends only on its magnitude

$$k = |\kappa| > 0, \quad (13)$$

because the horizontal second derivative gives

$$\partial_{xx} [\hat{\phi}_{j,\kappa}(z, t) e^{i\kappa x}] = -\kappa^2 \hat{\phi}_{j,\kappa}(z, t) e^{i\kappa x} = -k^2 \hat{\phi}_{j,\kappa}(z, t) e^{i\kappa x}. \quad (14)$$

Substitution into Laplace's equation (8) therefore gives the same ordinary differential equation in both fluids:

$$\frac{d^2 \hat{\phi}_{j,\kappa}}{dz^2} - k^2 \hat{\phi}_{j,\kappa} = 0. \quad (15)$$

The two independent solutions are $\cosh(kz)$ and $\sinh(kz)$, or equivalently shifted versions of these functions. The shifted form is convenient because each wall condition can be imposed at the point where the shifted coordinate is zero.

For the upper fluid, introduce the distance from the upper wall,

$$y_1 = H_1 - z. \quad (16)$$

The most general modal potential may be written

$$\hat{\phi}_{1,\kappa}(z, t) = A_1(t) \cosh(k(H_1 - z)) + B_1(t) \sinh(k(H_1 - z)). \quad (17)$$

Differentiating with respect to z gives the upper vertical velocity amplitude

$$\widehat{w}_{1,\kappa}(z, t) = \partial_z \widehat{\phi}_{1,\kappa}(z, t) = -kA_1(t) \sinh(k(H_1 - z)) - kB_1(t) \cosh(k(H_1 - z)). \quad (18)$$

The rigid-wall condition at $z = H_1$ requires

$$\widehat{w}_{1,\kappa}(H_1, t) = -kB_1(t) = 0, \quad (19)$$

so $B_1(t) = 0$. Thus the upper potential has the reduced form

$$\widehat{\phi}_{1,\kappa}(z, t) = A_1(t) \cosh(k(H_1 - z)). \quad (20)$$

For the lower fluid, introduce the distance from the lower wall,

$$y_2 = z + H_2. \quad (21)$$

The most general modal potential compatible with the vertical ODE is

$$\widehat{\phi}_{2,\kappa}(z, t) = A_2(t) \cosh(k(z + H_2)) + B_2(t) \sinh(k(z + H_2)). \quad (22)$$

Its vertical derivative is

$$\widehat{w}_{2,\kappa}(z, t) = \partial_z \widehat{\phi}_{2,\kappa}(z, t) = kA_2(t) \sinh(k(z + H_2)) + kB_2(t) \cosh(k(z + H_2)). \quad (23)$$

The wall condition at $z = -H_2$ gives

$$\widehat{w}_{2,\kappa}(-H_2, t) = kB_2(t) = 0, \quad (24)$$

so $B_2(t) = 0$, and therefore

$$\widehat{\phi}_{2,\kappa}(z, t) = A_2(t) \cosh(k(z + H_2)). \quad (25)$$

It remains to normalize the two constants $A_1(t)$ and $A_2(t)$ by the interfacial kinematic condition. Define the modal interface velocity by

$$\widehat{W}(t) = \partial_t \widehat{\eta}(t), \quad (26)$$

where the subscript κ has been suppressed for readability. The linear kinematic condition (11) requires the vertical velocity on both sides of the interface to equal $\widehat{W}(t)$:

$$\widehat{w}_{1,\kappa}(0, t) = \widehat{w}_{2,\kappa}(0, t) = \widehat{W}(t). \quad (27)$$

For the upper fluid this gives

$$-kA_1(t) \sinh(kH_1) = \widehat{W}(t), \quad A_1(t) = -\frac{\widehat{W}(t)}{k \sinh(kH_1)}. \quad (28)$$

Hence

$$\widehat{\phi}_{1,\kappa}(z, t) = -\frac{\widehat{W}(t)}{k \sinh(kH_1)} \cosh(k(H_1 - z)), \quad (29)$$

and differentiating once more gives

$$\widehat{w}_{1,\kappa}(z, t) = \widehat{W}(t) \frac{\sinh(k(H_1 - z))}{\sinh(kH_1)}. \quad (30)$$

For the lower fluid,

$$kA_2(t) \sinh(kH_2) = \widehat{W}(t), \quad A_2(t) = \frac{\widehat{W}(t)}{k \sinh(kH_2)}. \quad (31)$$

Thus

$$\widehat{\phi}_{2,\kappa}(z, t) = \frac{\widehat{W}(t)}{k \sinh(kH_2)} \cosh(k(z + H_2)), \quad (32)$$

and

$$\widehat{w}_{2,\kappa}(z, t) = \widehat{W}(t) \frac{\sinh(k(z + H_2))}{\sinh(kH_2)}. \quad (33)$$

Equations (30) and (33) are the finite-depth eigenfunctions used in the notebook. They have three important built-in properties:

$$\widehat{w}_{1,\kappa}(H_1, t) = 0, \quad \widehat{w}_{2,\kappa}(-H_2, t) = 0, \quad \widehat{w}_{1,\kappa}(0, t) = \widehat{w}_{2,\kappa}(0, t) = \widehat{W}(t). \quad (34)$$

The same normalized potentials also give the interface values needed in the dynamic condition:

$$\begin{aligned} \widehat{\phi}_{1,\kappa}(0, t) &= -\frac{\widehat{W}(t)}{k} \frac{\cosh(kH_1)}{\sinh(kH_1)} = -\frac{\coth(kH_1)}{k} \partial_t \widehat{\eta}(t), \\ \widehat{\phi}_{2,\kappa}(0, t) &= \frac{\widehat{W}(t)}{k} \frac{\cosh(kH_2)}{\sinh(kH_2)} = \frac{\coth(kH_2)}{k} \partial_t \widehat{\eta}(t). \end{aligned} \quad (35)$$

The opposite signs in the two interface potentials come from the opposite directions from the interface to the two walls: in the upper fluid increasing z points toward the wall, while in the lower fluid increasing z points away from the lower wall. The factors $\rho_1 \coth(kH_1)$ and $\rho_2 \coth(kH_2)$ that appear in the dispersion relation come from these interface potential values. They represent the finite-depth inertial loading of the two fluids on the moving interface.

2.3 Dynamic condition and surface tension

The linearized unsteady Bernoulli relation gives the perturbation pressure in each fluid. Evaluated at the displaced interface and retaining the hydrostatic contribution from moving the interface by η , the pressure jump can be written as

$$p_1 - p_2 = -\rho_1 \partial_t \phi_1(x, 0, t) + \rho_2 \partial_t \phi_2(x, 0, t) - (\rho_1 - \rho_2) g \eta. \quad (36)$$

With the sign convention used in this note, the linearized capillary pressure jump is

$$p_1 - p_2 = \sigma \partial_{xx} \eta. \quad (37)$$

Equivalently,

$$\rho_1 \partial_t \phi_1(x, 0, t) - \rho_2 \partial_t \phi_2(x, 0, t) + (\rho_1 - \rho_2) g \eta = -\sigma \partial_{xx} \eta. \quad (38)$$

For a Fourier mode, $-\partial_{xx} \eta = k^2 \eta$. Substituting (35) into (38) gives

$$-\frac{\rho_1 \coth(kH_1) + \rho_2 \coth(kH_2)}{k} \partial_{tt} \widehat{\eta} + (\rho_1 - \rho_2) g \widehat{\eta} = \sigma k^2 \widehat{\eta}. \quad (39)$$

Solving for $\partial_{tt} \widehat{\eta}$ yields the modal growth coefficient used throughout the note.

2.4 Stability information in the dispersion relation

For $k > 0$ the denominator

$$D(k) = \rho_1 \coth(kH_1) + \rho_2 \coth(kH_2) \quad (40)$$

is positive. Hence stability is determined by the sign of the numerator

$$k(\rho_1 - \rho_2)g - \sigma k^3. \quad (41)$$

The gravitational term is destabilizing when the upper fluid is heavier than the lower fluid, $\rho_1 > \rho_2$. The capillary term is stabilizing and grows like k^3 , so sufficiently short waves oscillate rather than grow.

The finite-depth factors modify the effective inertia. In the deep-water limit $kH_1, kH_2 \gg 1$, $\coth(kH_j) \rightarrow 1$ and the denominator approaches $\rho_1 + \rho_2$. In the shallow, long-wave limit $kH_j \ll 1$, $\coth(kH_j) \sim 1/(kH_j)$, so the denominator becomes large and the growth of very long waves is reduced by the rigid boundaries.

2.5 Initial-value interpretation

The modal equation has the form

$$\frac{d^2 \hat{\eta}}{dt^2} = \lambda \hat{\eta}, \quad (42)$$

where $\lambda = \omega^2(k)$ in the convention of this note. The kernel $G(\lambda, t)$ used below is the Green's function for the initial data

$$G(\lambda, 0) = 0, \quad \partial_t G(\lambda, 0) = 1. \quad (43)$$

Therefore $\hat{\eta}(t) = \hat{v} G(\lambda, t)$ automatically enforces a flat initial interface and an imposed initial velocity $\hat{v}$. Positive λ gives hyperbolic growth, negative λ gives trigonometric oscillation, and the case $\lambda = 0$ is obtained by the continuous limit $G(0, t) = t$.

For a real initial velocity profile, the Fourier coefficients satisfy the conjugate symmetry

$$\hat{v}_{-n} = \overline{\hat{v}_n}. \quad (44)$$

The reconstructed interface is real for this reason; taking the real part in the implementation is a robust numerical way to remove tiny imaginary round-off residuals.

2.6 Discrete projection and verification

The notebook computes Fourier coefficients with a discrete periodic projection. This is the discrete Fourier transform normalized to match the continuous convention

$$\hat{v}_n = \frac{1}{L} \int_0^L v_0(x) e^{-ik_n x} dx. \quad (45)$$

For smooth periodic data, the periodic trapezoidal rule converges rapidly, and for trigonometric data whose modes are resolved by the grid it recovers the exact coefficients up to floating-point round-off.

The verification tests follow directly from the kernel and shape functions. Since $G(\lambda, 0) = 0$, one should have $\eta(x, 0) = 0$. Since $\partial_t G(\lambda, 0) = 1$, one should have $\partial_t \eta(x, 0) = v_0(x)$ after reconstruction. Finally, the hyperbolic sine factors in (30)–(33) vanish at $z = H_1$ and $z = -H_2$, so the wall velocity checks should be zero up to numerical round-off.

3 Finite-depth dispersion relation

For a Fourier mode with horizontal wavenumber $k > 0$, the corrected finite-depth dispersion relation used in the notebook is

$$\omega^2(k) = \frac{k(\rho_1 - \rho_2)g - \sigma k^3}{\rho_1 \coth(kH_1) + \rho_2 \coth(kH_2)}. \quad (46)$$

The sign of the capillary term is stabilizing: for sufficiently large k , the term $-\sigma k^3$ can make $\omega^2(k)$ negative, producing stable oscillatory modes rather than exponential growth.

When $\rho_1 > \rho_2$, the critical wavenumber at which the numerator changes sign is

$$k_c = \sqrt{\frac{(\rho_1 - \rho_2)g}{\sigma}}. \quad (47)$$

For $0 < k < k_c$, the numerator of (46) is positive, and the corresponding modes are Rayleigh–Taylor unstable. For $k > k_c$, surface tension stabilizes the mode in the linear theory.

4 Adapted normal-mode result from the shared derivation

The shared derivation uses a different but closely related linear method. It assumes semi-infinite viscous fluids and writes each perturbation as a normal mode,

$$f'(x, z, t) = \tilde{f}(z)e^{i\kappa x + st}, \quad k = |\kappa|. \quad (48)$$

Here s is a temporal growth rate obtained from a dispersion relation. To avoid conflicting with the indexing convention of this note, denote the upper semi-infinite fluid by u and the lower semi-infinite fluid by ℓ in this section.

4.1 Viscous semi-infinite vertical velocities

In the shared method, the bulk equations are the linearized incompressible Navier–Stokes equations,

$$\nabla \cdot \mathbf{U}'_j = 0, \quad \partial_t \mathbf{U}'_j = -\frac{1}{\rho_j} \nabla P'_j + \nu_j \nabla^2 \mathbf{U}'_j, \quad \nu_j = \frac{\mu_j}{\rho_j}. \quad (49)$$

After substituting (48), the derivative rules are

$$\partial_t \mapsto s, \quad \partial_x \mapsto i\kappa, \quad \nabla^2 \mapsto D^2 - k^2, \quad D = \frac{d}{dz}. \quad (50)$$

Incompressibility gives

$$i\kappa \tilde{U}_{jx} + D \tilde{U}_{jz} = 0. \quad (51)$$

Taking the curl of the linearized momentum equation removes the pressure and gives the vorticity-diffusion equation. Written only in terms of the vertical velocity amplitude, this is

$$[s - \nu_j(D^2 - k^2)](D^2 - k^2)\tilde{U}_{jz} = 0. \quad (52)$$

Thus the characteristic roots are

$$r = \pm k, \quad r = \pm q_j, \quad q_j^2 = k^2 + \frac{s}{\nu_j}. \quad (53)$$

The $r = \pm k$ roots are the harmonic, potential-flow part; the $r = \pm q_j$ roots are the viscous vorticity-diffusion part.

For semi-infinite fluids, the perturbations must decay away from the interface. If the lower fluid occupies $z < 0$ and the upper fluid occupies $z > 0$, the admissible vertical velocities are

$$U'_{\ell z} = (A_\ell e^{kz} + A'_\ell e^{q_\ell z}) e^{i\kappa x + st}, \quad z < 0, \quad (54)$$

$$U'_{uz} = (A_u e^{-kz} + A'_u e^{-q_u z}) e^{i\kappa x + st}, \quad z > 0. \quad (55)$$

The amplitudes are not fixed by the bulk equations alone. They are determined by interfacial kinematic, velocity-continuity, stress, and any additional physical jump conditions. The condition for a nonzero amplitude vector gives the dispersion relation for s .

4.2 Single-mode interface evolution

The shared derivation also assumes the interface has the same normal-mode dependence,

$$H(x, t) = Be^{i\kappa x + st}. \quad (56)$$

In real form, for a cosine perturbation,

$$H(x, t) = ae^{st} \cos(kx). \quad (57)$$

The linearized kinematic condition at $z = 0$ gives

$$\partial_t H = U'_{\ell z} = U'_{uz}. \quad (58)$$

For example, from the lower side,

$$sB = A_\ell + A'_\ell, \quad B = \frac{A_\ell + A'_\ell}{s}. \quad (59)$$

Thus the amplitude a in (57) is the initial displacement amplitude:

$$H(x, 0) = a \cos(kx). \quad (60)$$

If the spike height is evaluated at $x = \lambda/2$, where $k = 2\pi/\lambda$, then

$$H(\lambda/2, t) = ae^{st} \cos(\pi) = -ae^{st}. \quad (61)$$

4.3 How the two methods differ

The finite-depth notebook method and the shared normal-mode method are consistent in their common inviscid, deep-water limit, but they solve different linear problems.

- **Bulk model.** The present note assumes inviscid, irrotational flow, so each velocity potential satisfies Laplace's equation. The shared derivation uses viscous linearized Navier–Stokes equations. This adds the q_j roots in (53), which are absent from the potential-flow note.
- **Vertical structure.** The present note uses finite-depth eigenfunctions satisfying rigid-wall impermeability:

$$\frac{\sinh(k(H_1 - z))}{\sinh(kH_1)}, \quad \frac{\sinh(k(z + H_2))}{\sinh(kH_2)}. \quad (62)$$

The shared derivation uses semi-infinite decay conditions and therefore obtains exponentials. In the deep-water limit,

$$\frac{\sinh(k(H_1 - z))}{\sinh(kH_1)} \rightarrow e^{-kz}, \quad \frac{\sinh(k(z + H_2))}{\sinh(kH_2)} \rightarrow e^{kz}, \quad (63)$$

so the $e^{\mp kz}$ pieces agree with the potential-flow part of the shared method.

- **How time dependence is obtained.** The present note first derives $\lambda = \omega^2(k)$ and then solves the second-order modal initial-value problem

$$\frac{d^2 \hat{\eta}}{dt^2} = \lambda \hat{\eta}. \quad (64)$$

The shared derivation assumes the time dependence e^{st} from the beginning and finds admissible s from a dispersion relation.

- **Initial data.** The notebook problem imposes a flat initial interface and a prescribed initial interface velocity,

$$\hat{\eta}(0) = 0, \quad \partial_t \hat{\eta}(0) = \hat{v}. \quad (65)$$

For an unstable mode with $\lambda = s^2 > 0$, this gives

$$\hat{\eta}(t) = \frac{\hat{v}}{s} \sinh(st) = \frac{\hat{v}}{2s} (e^{st} - e^{-st}). \quad (66)$$

The shared result instead prescribes an initial displacement and keeps the growing normal-mode branch,

$$H(x, 0) = a \cos(kx), \quad H(x, t) = ae^{st} \cos(kx). \quad (67)$$

In the second-order notation, this corresponds to initial data

$$\hat{\eta}(0) = a, \quad \partial_t \hat{\eta}(0) = sa, \quad (68)$$

which removes the decaying branch.

- **Dispersion relation.** The present note has the explicit inviscid finite-depth formula (46). The shared method obtains s from a determinant condition involving the viscous amplitudes, stress balances, and any extra interfacial physics. Its dispersion relation is therefore generally more complicated and depends on viscosities through q_j .

5 Modal initial-value problem

Consider a periodic domain of length L . The horizontal wavenumbers are

$$k_n = \frac{2\pi n}{L}, \quad n \in \mathbb{Z}. \quad (69)$$

The zero mode is omitted in the notebook, since the instability calculation concerns nonzero interfacial perturbations. The initial velocity is expanded as

$$v_0(x) = \sum_{n \neq 0} \hat{v}_n e^{ik_n x}, \quad (70)$$

with Fourier coefficients

$$\hat{v}_n = \frac{1}{L} \int_0^L v_0(x) e^{-ik_n x} dx. \quad (71)$$

For each nonzero mode, define

$$\lambda_n = \omega^2(|k_n|). \quad (72)$$

The modal interface amplitude satisfies

$$\frac{d^2 \hat{\eta}_n}{dt^2} = \lambda_n \hat{\eta}_n, \quad (73)$$

with initial conditions

$$\hat{\eta}_n(0) = 0, \quad \frac{d\hat{\eta}_n}{dt}(0) = \hat{v}_n. \quad (74)$$

The solution is

$$\hat{\eta}_n(t) = \hat{v}_n G(\lambda_n, t), \quad (75)$$

where

$$G(\lambda, t) = \begin{cases} \frac{\sinh(\sqrt{\lambda} t)}{\sqrt{\lambda}}, & \lambda \neq 0, \\ t, & \lambda = 0. \end{cases} \quad (76)$$

Its time derivative is

$$\partial_t G(\lambda, t) = \begin{cases} \cosh(\sqrt{\lambda} t), & \lambda \neq 0, \\ 1, & \lambda = 0. \end{cases} \quad (77)$$

If $\lambda < 0$, these formulae are interpreted through complex arguments. Equivalently, writing $\lambda = -\alpha^2$ with $\alpha > 0$, one obtains

$$G(-\alpha^2, t) = \frac{\sin(\alpha t)}{\alpha}, \quad \partial_t G(-\alpha^2, t) = \cos(\alpha t), \quad (78)$$

which corresponds to stable capillary-gravity oscillations.

The reconstructed interface displacement is

$$\eta(x, t) = \text{Re} \left[\sum_{n \neq 0} \hat{\eta}_n(t) e^{ik_n x} \right]. \quad (79)$$

The interfacial vertical velocity is

$$w(x, 0, t) = \partial_t \eta(x, t) = \text{Re} \left[\sum_{n \neq 0} \hat{v}_n \partial_t G(\lambda_n, t) e^{ik_n x} \right]. \quad (80)$$

6 Vertical velocity fields

The finite-depth vertical structure used in the notebook satisfies the impermeability conditions at the two walls. If the interfacial vertical velocity amplitude of mode n is denoted by

$$\hat{w}_n(0, t) = \hat{v}_n \partial_t G(\lambda_n, t), \quad (81)$$

then the upper-fluid vertical velocity is reconstructed using

$$w_{1,n}(z, t) = \hat{w}_n(0, t) \frac{\sinh(|k_n|(H_1 - z))}{\sinh(|k_n|H_1)}. \quad (82)$$

Similarly, the lower-fluid vertical velocity is

$$w_{2,n}(z, t) = \hat{w}_n(0, t) \frac{\sinh(|k_n|(z + H_2))}{\sinh(|k_n|H_2)}. \quad (83)$$

Therefore,

$$w_1(x, z, t) = \text{Re} \left[\sum_{n \neq 0} \hat{w}_n(0, t) \frac{\sinh(|k_n|(H_1 - z))}{\sinh(|k_n|H_1)} e^{ik_n x} \right], \quad (84)$$

for $0 < z < H_1$, and

$$w_2(x, z, t) = \text{Re} \left[\sum_{n \neq 0} \hat{w}_n(0, t) \frac{\sinh(|k_n|(z + H_2))}{\sinh(|k_n|H_2)} e^{ik_n x} \right], \quad (85)$$

for $-H_2 < z < 0$.

These formulae immediately imply

$$w_1(x, H_1, t) = 0, \quad w_2(x, -H_2, t) = 0, \quad (86)$$

and

$$w_1(x, 0, t) = w_2(x, 0, t) = \partial_t \eta(x, t). \quad (87)$$

7 Example used in the notebook

The notebook uses the parameter set

$$\rho_1 = 2, \quad \rho_2 = 1, \quad g = 9.81, \quad \sigma = 0.01, \quad H_1 = H_2 = 1, \quad L = 2\pi. \quad (88)$$

The imposed initial interfacial velocity is

$$v_0(x) = U_0 \cos(3x) + \frac{2}{5}U_0 \sin(5x), \quad U_0 = 10^{-3}. \quad (89)$$

Since $L = 2\pi$, the nonzero exact Fourier coefficients are

$$\hat{v}_3 = \frac{U_0}{2}, \quad \hat{v}_{-3} = \frac{U_0}{2}, \quad (90)$$

$$\hat{v}_5 = -\frac{iU_0}{5}, \quad \hat{v}_{-5} = \frac{iU_0}{5}. \quad (91)$$

The numerical notebook nevertheless computes the coefficients through a discrete periodic projection, which is more general and can be used for arbitrary smooth periodic profiles.

8 Discrete Fourier projection in the Mathematica notebook

The notebook uses a uniform grid

$$x_m = \frac{mL}{N}, \quad m = 0, 1, \dots, N-1, \quad (92)$$

where $N = 4096$. The Fourier coefficient is approximated by the periodic trapezoidal rule

$$\hat{v}_n \approx \frac{1}{N} \sum_{m=0}^{N-1} v_0(x_m) e^{-ik_n x_m}. \quad (93)$$

This avoids the convergence warnings caused by numerical quadrature when the exact coefficient is zero. The Mathematica implementation also applies **Chop** to remove round-off residuals.

9 Verification conditions

The notebook verifies the initial conditions by computing

$$\max_x |\eta(x, 0)|, \quad \max_x |\partial_t \eta(x, 0) - v_0(x)|. \quad (94)$$

For a sufficiently resolved Fourier truncation, both quantities should be close to machine precision.

The wall boundary conditions are verified by computing

$$\max_x |w_1(x, H_1, 0)|, \quad \max_x |w_2(x, -H_2, 0)|. \quad (95)$$

Because the vertical shape functions vanish exactly at the corresponding rigid walls, these errors should also be at the level of numerical round-off.

A Core Mathematica implementation

The following listing contains the essential part of the corrected Mathematica notebook.

```
Remove["Global`*"];

pars = {
  rho1 -> 2.0,
  rho2 -> 1.0,
  g -> 9.81,
  sigma -> 0.01,
  H1 -> 1.0,
  H2 -> 1.0,
  L -> 2 Pi
};

Lnum = N[L /. pars];

omega2[k_] := (k (rho1 - rho2) g - sigma k^3)/
  (rho1 Coth[k H1] + rho2 Coth[k H2]);

omega2Num[k_?NumericQ] := N[omega2[k] /. pars];

kCritical = Sqrt[((rho1 - rho2) g)/sigma] /. pars;

tol = 10^-12;

growthKernel[lambda_?NumericQ, t_?NumericQ] :=
  If[Abs[lambda] < tol,
    t,
    Sinh[Sqrt[lambda] t]/Sqrt[lambda]
  ];

growthKernelDot[lambda_?NumericQ, t_?NumericQ] :=
  If[Abs[lambda] < tol,
    1,
    Cosh[Sqrt[lambda] t]
  ];

U0 = 10^-3;

v0Expr = U0 Cos[3 x] + (2/5) U0 Sin[5 x];

v0[xVal_?NumericQ] := N[v0Expr /. x -> xVal];

nMax = 32;

kn[n_Integer] := 2 Pi n/Lnum;

modes = Join[Range[-nMax, -1], Range[1, nMax]];

nSamples = 4096;

xGrid = N[Lnum Range[0, nSamples - 1]/nSamples];
v0Grid = v0 /@ xGrid;

Clear[vHat];

vHat[n_Integer] := vHat[n] =
```

```

Chop[
  (1/nSamples) Total[v0Grid Exp[-I kn[n] xGrid]],
  10^-14
];

etaHat[n_Integer, tVal_?NumericQ] :=
  vHat[n] growthKernel[omega2Num[Abs[kn[n]]], tVal];

eta[xVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      etaHat[#, tVal] Exp[I kn[#] xVal] & /@ modes
    ]
  ];

wInterfaceHat[n_Integer, tVal_?NumericQ] :=
  vHat[n] growthKernelDot[omega2Num[Abs[kn[n]]], tVal];

wInterface[xVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      wInterfaceHat[#, tVal] Exp[I kn[#] xVal] & /@ modes
    ]
  ];

shape1[k_?NumericQ, zVal_?NumericQ] :=
  Sinh[k ((H1 /. pars) - zVal)]/Sinh[k (H1 /. pars)];

shape2[k_?NumericQ, zVal_?NumericQ] :=
  Sinh[k (zVal + (H2 /. pars))]/Sinh[k (H2 /. pars)];

w1[xVal_?NumericQ, zVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      (wInterfaceHat[#, tVal]
       shape1[Abs[kn[#]], zVal]
       Exp[I kn[#] xVal]) & /@ modes
    ]
  ];

w2[xVal_?NumericQ, zVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      (wInterfaceHat[#, tVal]
       shape2[Abs[kn[#]], zVal]
       Exp[I kn[#] xVal]) & /@ modes
    ]
  ];

checkGrid = N[Lnum Range[0, 512]/512];

maxEtaInitialError =
  Max[Abs[eta[#, 0] & /@ checkGrid]];

maxVelocityInitialError =
  Max[Abs[(wInterface[#, 0] - v0[#]) & /@ checkGrid]];

maxUpperWallError =

```

```
Max[Abs[w1[#, H1 /. pars, 0] & /@ checkGrid]];

maxLowerWallError =
  Max[Abs[w2[#, -(H2 /. pars), 0] & /@ checkGrid]];

{
  maxEtaInitialError,
  maxVelocityInitialError,
  maxUpperWallError,
  maxLowerWallError
}
```